Final Conclusion Report: Full-Process Destructive Test Summary Report

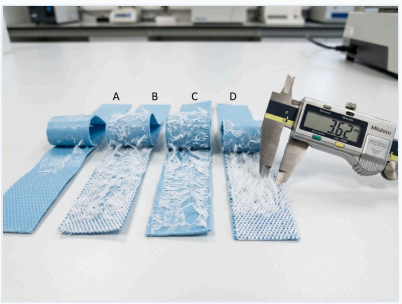
Consolidated Verification Findings Across Stage 1 (Material), Stage 2 (Heat-Welding & Adhesive Process), and Stage 3 (Structural Design)

1. Product Technical Specifications & Full-Process Traceability

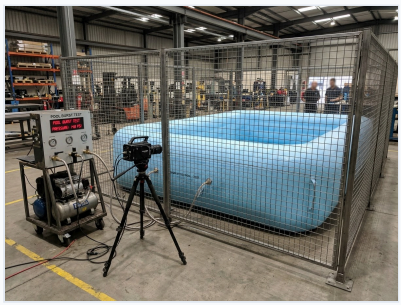
Final Report No.	`RPT-FB-QC-20261113-FINAL`	**Linked Sub-Reports**	`S1 / S2 / S3 (per WI-QC-DS-2026-001)`
Brand / Model	**Felizbay** Large Rectangular Drop-Stitch Pool (`FB-DSP-400`)	**Dimensions / Wall**	**400 cm × 300 cm × 120 cm** / Wall Thickness **20 cm**
Bonding Process	**RF / Hot-Air Heat Welding + 2-Part PU Cold-Bonding Adhesive**	**Rated Load**	**P_w = 8.0 PSI** (0.55 bar) / Rated Water Volume 9,828 L
Batch & Samples	Batch `LOT-202611-A08` (**Units #1, #2, #3** — 3 Complete Units)	**Material / Glue Lots**	Roll `R261029-04B` / Tape `WB-261015-02` / Glue `GL-20261022`
Test Period	2026-11-09 to 2026-11-13 (Incl. 48 h Thermal Creep + 72 h Hydrostatic)	**Final Disposition**	**PASS (All 8 Destructive Items 100% Compliant — Batch Released)**

Executive Final Determination (PASS — Approved for Mass-Production Shipment): The **Felizbay** Large Rectangular Drop-Stitch Inflatable Pool (Model `FB-DSP-400`, **400 cm × 300 cm × 120 cm**, **20 cm** wall thickness, manufactured via **RF/hot-air heat welding + two-component polyurethane cold-bonding adhesive**) has completed all 8 destructive verification tests across Stage 1 (Material Properties), Stage 2 (Bonding Process Limits), and Stage 3 (Structural Design). Every test parameter passed factory specifications with wide safety margins; zero material, process, or design defects were identified.

2. Photographic Evidence Summary Across All Three Stages



Stage 1 Evidence (RPT-S1)
1000D drop-stitch fabric exhibited 100% cohesive fiber tear (mean 136.5–149.4 N/5cm), confirming superior PVC coating anchorage.



Stage 2 Evidence (RPT-S2)
400 × 300 × 120 cm unit reached 25.4 PSI (3.18×P_w) burst in main panel fabric with zero failure in heat-welded or PU-bonded seams.



Stage 3 Evidence (RPT-S3)
20 cm thick wall held 110% water overload (10,810 L) for 72 h at only 5.6° tilt and survived 500 pendulum impacts with zero leaks.

3. Consolidated 8-Item Full-Process Destructive Test Matrix

Test Stage	Test Sub-Item	Unit	Factory Acceptance Spec	Felizbay Measured Result	Result
Stage 1 Material (RPT-S1)	1.1 Coating 180° Peel Test	Unit #1	Mean ≥ 120 / Trough ≥ 100 N/5cm	Mean 136.5–149.4 / Min 115.4 N/5cm	**PASS**
	1.2 Pressurized Puncture Test	Unit #1	Force ≥ 350 N; Tear ≤ 30 mm	Force 406.2–425.0 N; Tear ≤ 18.2 mm	**PASS**
	1.3 -18°C Cold Folding Test	Unit #1	Zero cracking; 3 min 0 bubbles	All 6 panel & seam cuts 0 bubbles	**PASS**
Stage 2 Process (RPT-S2)	2.1 55°C / 1.5P_w 48h Creep	Unit #2	Expansion ≤ 10%; Slip ≤ 1.0 mm	Max expansion 5.2%; Max slip 0.42 mm	**PASS**
	2.2 Step-Up Overpressure Burst	Unit #2	P_{burst} ≥ 2.5 × P_w (20.0 PSI)	P_{burst} = **25.4 PSI (3.18 × P_w)**, panel burst	**PASS**
Stage 3 Structure (RPT-S3)	3.1 110% Overload Water 72h	Unit #3	Tilt θ ≤ 10°; zero inner seam peel	Max tilt 5.6°; inner seam & gusset intact	**PASS**
	3.2 150 kg Load & 500 Impacts	Unit #3	No buckling; 1.5P_w 1 h 0 leak	11.8 mm sag (no buckle); 0.0 PSI drop	**PASS**
	3.3 Valve Torque & Drain Pull	Unit #3	Torque ≥ 25 N·m; Pull ≥ 800 N	Torque 31.9–36.4 N·m; Pull 1,185 N	**PASS**

4. Technical Dimension Assessment & Mass-Production Control Plan

- **Raw Material Integrity (Stage 1)**: The 1000D high-density drop-stitch fabric exhibits strong resin-to-yarn mechanical anchorage, > 16% puncture force margin under 8.0 PSI inflation, and stable -18°C cold-flexibility, fully supporting large deep-water basin tension.
- **Heat-Welding & Polyurethane Adhesive Process (Stage 2)**: Combining RF/hot-air welding on linear edge bands with two-component PU cold-bonding on 3D corners and fitting flanges yielded only 0.42 mm seam slip after 48 h at 55°C / 12.0 PSI and withstood 3.18×P_w burst pressure without seam separation.
- **Deep-Water Structural Design (Stage 3)**: For the 400 × 300 × 120 cm capacity (10,810 L overload), the **20 cm thick drop-stitch wall** provides ample section modulus (5.6° max tilt over 72 h), while the internal perimeter triangular gusset and R160 mm double-layered corner patches eliminate corner stress concentrations.
- **Production Quality Control Recommendation (CAPA Preventive Control)**: Maintain per-shift first-article peel checks on welded edge bands (≥ 120 N/5cm) and enforce strict two-component PU adhesive mixing ratios (100:5) and open-time window monitoring during assembly.

Lab Supervisor: H. Chen (QC Lab)	**QA Manager**: M. Lin (Quality Eng.)	**Approved for Release**: W. Zhou (Quality Director)